Young Kyung Kim

AI alignment · Interpretability · Monitorability · Test-time adaptation · Computational health

yk4491@princeton.edu | youngkyung.kim@duke.edu | Google Scholar | GitHub: youngkyungkim93

I develop **interpretable and monitorable AI systems to support human oversight and advance AI alignment**. My research asks: **what does monitorability mean beyond language?** I introduced Chain-of-Image Generation (CoIG), to our knowledge the first systematic framework for monitorable image generation, and am extending this work to 3D generation. I also study how models can learn from the order of incoming data at test time, alongside interpretability, structured reasoning, and clinical research on eating disorders and mental health.

RESEARCH APPOINTMENT AND EDUCATION

Princeton University | Visiting research appointment — 2024 - Present

Electrical and Computer Engineering · Guillermo Sapiro research group

Duke University | Ph.D. candidate, Electrical and Computer Engineering — 2021 - Present

Research advisor: Guillermo Sapiro

Associate Member, Rhodes Information Initiative (Rhodes iiD) · 2021 - Present

Duke University | Master of Interdisciplinary Data Science — 2021

SELECTED RESEARCH CONTRIBUTIONS

Monitorable generation and structured reasoning

CoIG · arXiv 2025 | CLARO · Findings of ACL 2026 | 3D generation · ongoing

- **CoIG:** developed image generation through inspectable semantic steps; introduced readability and causal-relevance metrics to assess how intermediate steps reveal and influence the final image.
- **CLARO:** developed a reinforcement learning framework that improves reasoning readability and efficiency by optimizing structural attributes within a token budget.
- **Ongoing 3D research:** extending monitorability to 3D generation to support human oversight and intervention.

Interpretability

MNAM · TMLR 2024 | Semantic Vision Transformers · arXiv 2024

- **MNAM:** developed mixtures of neural additive models that capture multiple statistical relationships while preserving interpretable feature-response functions.
- **sViT:** introduced segmentation-based semantic image tokens to make visual representations more interpretable and improve robustness to distribution shifts.

Learning from sequence order at test time

Order-Aware Test-Time Adaptation (OATTA) · arXiv 2026

- Developed OATTA to learn from the order of incoming data, estimating changing class-transition probabilities online from unlabeled model predictions.
- To our knowledge, the first general TTA framework to exploit latent class-transition matrices; refines predictions through Bayesian filtering without updating model weights.

Computational research in eating disorders and mental health

Clinical collaborations · Duke University

- Analyzed feeding strategies from 19,239 adults; NLP classified perceived helpfulness with 82% accuracy.
- Studied facial behavior during ARFID food exposures; co-authored college mental health peer-support research.

PUBLICATIONS AND PREPRINTS

Newest first within each group · 11 distinct works

First-author publications and preprints

1. Order-Aware Test-Time Adaptation: Leveraging Temporal Dynamics for Robust Streaming Inference
Young Kyung Kim, Oded Schlesinger, Qiangqiang Wu, J. Matías Di Martino, and Guillermo Sapiro. *arXiv preprint arXiv:2601.21012, 2026.*

2. Chain-of-Image Generation: Toward Monitorable and Controllable Image Generation
Young Kyung Kim, Oded Schlesinger, Yuzhou Zhao, J. Matias Di Martino, and Guillermo Sapiro. *arXiv preprint arXiv:2512.08645, 2025.*

3. Vision Transformers with Natural Language Semantics
Young Kyung Kim, J. Matías Di Martino, and Guillermo Sapiro. *arXiv preprint arXiv:2402.17863, 2024.*

4. Generalizing Neural Additive Models via Statistical Multimodal Analysis
Young Kyung Kim, J. Matías Di Martino, and Guillermo Sapiro. *Transactions on Machine Learning Research, 2024.*

5. A Tale of Two Food Adventurers: The Challenges and Triumphs of Repeated Food Exposures in Avoidant/Restrictive Food Intake Disorder
Young Kyung Kim, Juan Matias Di Martino, Julia Nicholas, Ilana Pilato, Alannah Melissa Rivera-Cancel, Julia Gianneschi, Jalisa Jackson, Ellen Vernell Mines, Nancy Zucker, and Guillermo Sapiro. *Empowering Communities: A Participatory Approach to AI for Mental Health (PAI4MH), NeurIPS 2022 Workshop.*

6. Parent strategies for expanding food variety: Reflections of 19,239 adults with symptoms of Avoidant/Restrictive Food Intake Disorder
Young Kyung Kim, J. Matías Di Martino, Julia Nicholas, Alannah Rivera-Cancel, Jennifer E. Wildes, Marsha D. Marcus, Guillermo Sapiro, and Nancy Zucker. *International Journal of Eating Disorders, 55(1):108-119, 2022.*

Additional co-authored publications and preprints

7. An Evaluation Framework for Generating Multi-View Images of a Person in a Scene
Mahir Majid, **Young Kyung Kim**, and Guillermo Sapiro. *arXiv preprint arXiv:2609.04603, 2026.*

8. CLARO: Controlled Attribute-Driven Reasoning Optimization for Efficient Chain-of-Thought
Oded Schlesinger, **Young Kyung Kim**, J. Matias Di Martino, and Guillermo Sapiro. *Findings of the Association for Computational Linguistics: ACL 2026, pp. 26779-26799, 2026.*

9. Peer support: Current status and future opportunities for college mental health promotion
Savannah Erwin, Talita Ahmed, Jonathan Cloughesy, Kyra Citron, Faith Joo, Elizabeth Monahan, **Young Kyung Kim**, Brenda Straka, Gary Bennett, Timothy Strauman, Sarah Gaither, Guillermo Sapiro, and Nancy Zucker. *Journal of American College Health, 74(2):413-423, 2026.*

10. Assessing Fears of Negative Consequences in Children With Symptoms of Avoidant Restrictive Food Intake Disorder
Julia R. Gianneschi, Kara A. Washington, Julia Nicholas, Ilana Pilato, Sarah LeMay-Russell, Alannah M. Rivera-Cancel, Ellen V. Mines, Jalisa E. Jackson, Samuel Marsan, Sage Lachman, **Young Kyung Kim**, J. Matias Di Martino, Jane Pendergast, Katharine L. Loeb, Debra K. Katzman, Marsha D. Marcus, Rachel Bryant-Waugh, Guillermo Sapiro, and Nancy L. Zucker. *International Journal of Eating Disorders, 57(12):2329-2340, 2024.*

11. Feeling and Body Investigators (FBI): An Interoceptive Exposure Treatment Approach for Young Children With Avoidant/Restrictive Food Intake Disorder (ARFID)
Kara A. Washington, Elizabeth M. Monahan, Faith Joo, Ilana Pilato, Alannah M. Rivera-Cancel, **Young Kyung Kim**, Eli Rotondo, J. Matias Di Martino, Valerie Smith, Katharine L. Loeb, Debra K. Katzman, Marsha Marcus, Rachel Bryant-Waugh, Guillermo Sapiro, and Nancy Zucker. *Conference abstract. Journal of the American Academy of Child & Adolescent Psychiatry, 63(10 Suppl.):S163, 2024. Also presented as poster 1.17 at AACAP's 2024 Annual Meeting.*

Technical skills: PyTorch · Model training · Evaluation design · Experimental analysis